

The Thermodynamic Origin of Quantization

Deriving Planck's Constant from a Hydrodynamic
Vacuum in Swirl-String Theory (SST)

A central claim: Quantization is not a fundamental postulate, but an emergent thermodynamic property.

Swirl-String Theory (SST) models matter as closed circulation-carrying filaments in an inviscid, incompressible vacuum-like medium.

This presentation outlines how discretization arises naturally from a topological circulation invariant combined with a Clausius-consistent work/heat decomposition.

Guiding Principle: The Zero-Free-Parameter Program. Our goal is to derive fundamental constants like h from a minimal set of mechanical inputs, without ad-hoc tuning.

A Taxonomy of Claims: Establishing a Framework for Rigor



Postulates

Foundational assumptions about the medium and its invariants.

The “Swirl Medium” is inviscid and incompressible.

Topological circulation, $\Gamma \equiv \oint \mathbf{v} \cdot d\vec{l}$, is an invariant.



Derivations

Results that follow logically from the postulates.

The thermodynamic mapping from mechanics to entropy.

The numerical values of fundamental constants (A_C , h).

Predicted scaling laws for heat capacity.



Speculations

Consistent hypotheses requiring further validation.

The “Golden Layer” hypothesis for mass selection.

The Foundation: A Minimal Primitive Set of Mechanical Inputs

The entire theory is built upon a normalized triad of primitive mechanical parameters.

Medium Density (ρ_f)

The effective inertial density of the far-field swirl medium.

$$7.0 \times 10^{-7} \text{ kg m}^{-3}$$

Core Radius (r_c)

The characteristic radius of the swirl-string filament core.

$$1.40897017 \times 10^{-15} \text{ m}$$

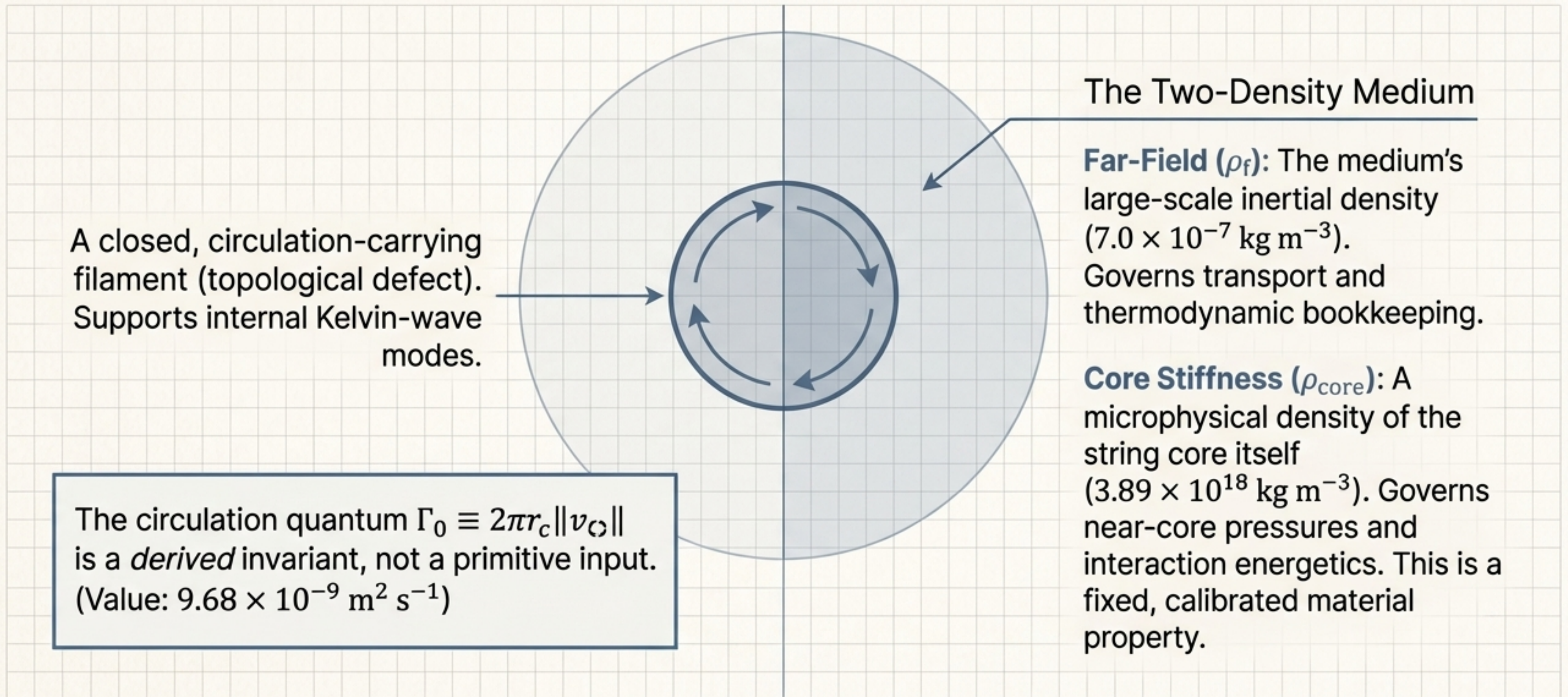
Swirl Speed ($\|\mathbf{v}_c\|$)

The characteristic tangential speed at the filament core.

$$1.09384563 \times 10^6 \text{ m s}^{-1}$$

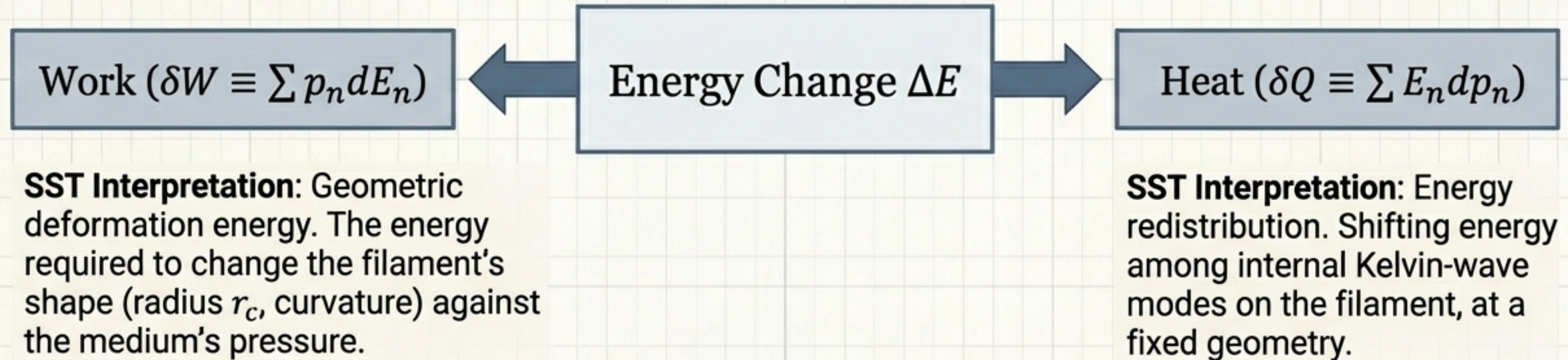
These calibrated inputs are the only 'dials' in the theory. All other constants are derived.

The Physical Model: Swirl Strings in a Two-Scale Medium



A Mechanical Theory of Heat and the Emergence of Discretization

Quantization emerges from how energy is partitioned in the filament. The Abe-Okuyama mapping provides a formal bridge from pure-state mechanics to quantum thermodynamics via the Clausius equality ($\oint \delta Q/T = 0$).



Key Insight: This decomposition provides a mechanical basis for the concepts of 'level shift' (Work) and 'population change' (Heat), forming the bedrock for an emergent quantum-like structure.

Step 1: Deriving the Interaction Coupling Scale (A_C)

The strength of the interaction between two swirl strings is not a new parameter but is derived from the core's mechanical properties. It is a near-core pressure invariant.

$$A(\text{SST})_C = 4\pi \cdot \rho_{\text{core}} \cdot \|\mathbf{v}_{\ominus}\|^2 \cdot r_c^4$$

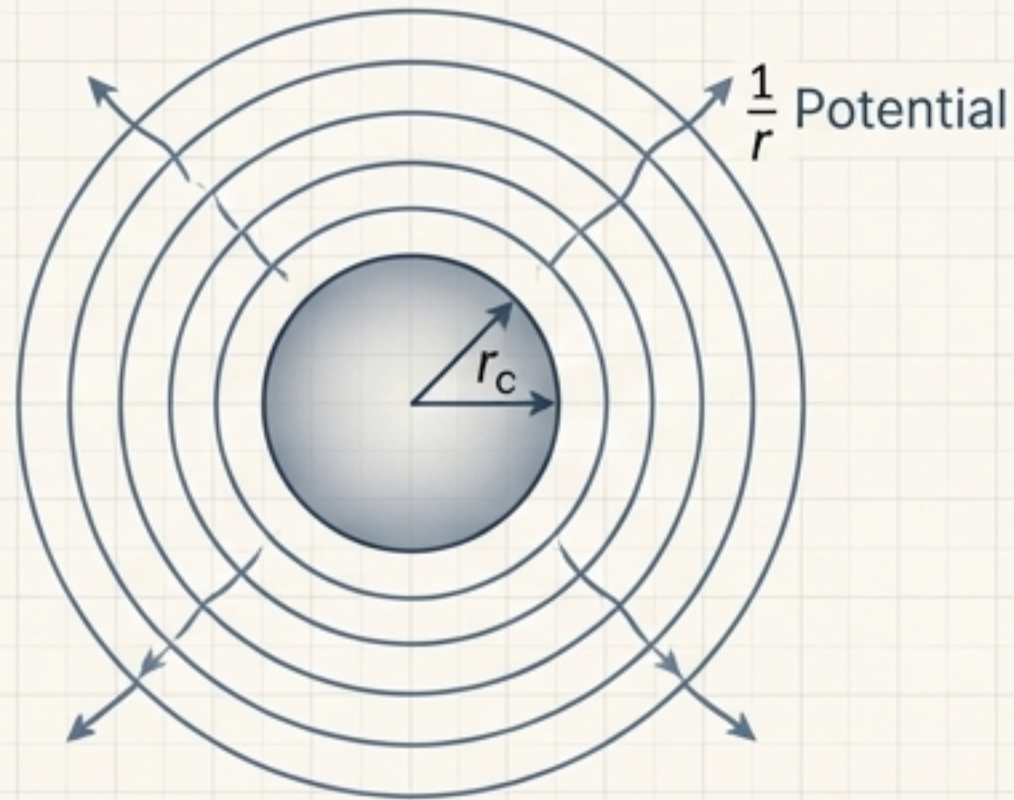
Derived Value: $A_C^{(\text{SST})} = 2.307077328 \times 10^{-28} \text{ J m}$

Benchmark: This numerically matches the empirical SI Coulomb coupling scale, $A_C^{(\text{SI})} \equiv e^2/4\pi\epsilon_0$. This is a parameter-free prediction.

Justifying the Geometry: The Origin of the 4π Prefactor

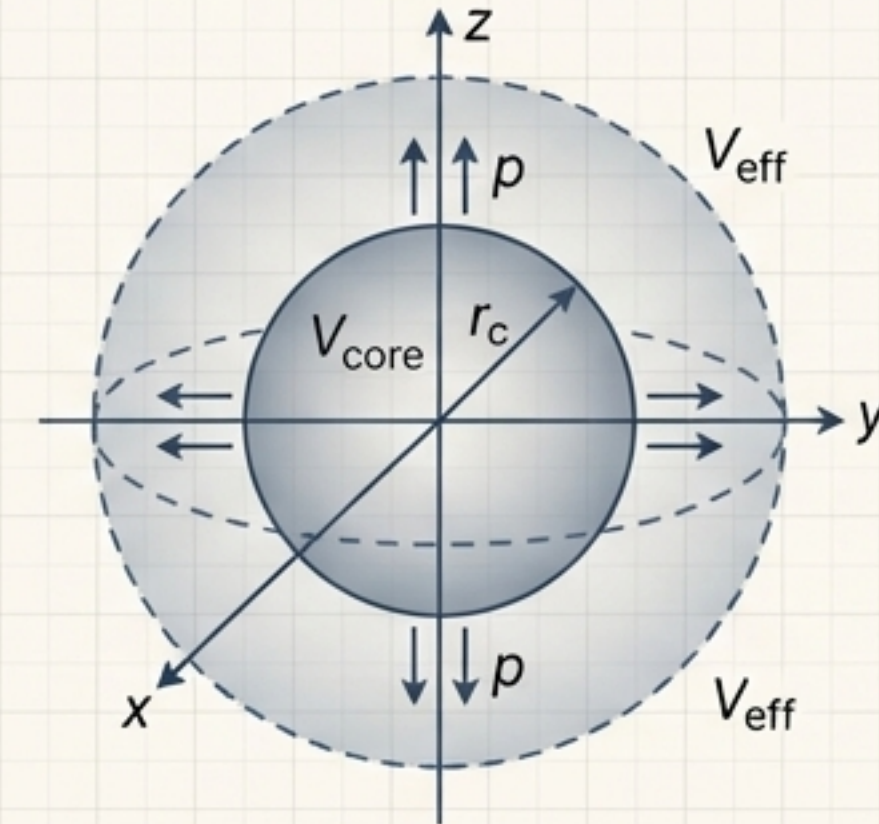
The ' 4π ' in ' A_C ' is not arbitrary. It is fixed by two standard normalizations in continuum mechanics and potential theory.

Part 1: Far-Field Potential (Laplace Uniqueness)



Outside the core ($r > r_c$), the static pressure perturbation δp obeys $\nabla^2 \delta p = 0$. The unique decaying, spherically symmetric solution is $\delta p(r) \propto \frac{1}{r}$. The 4π arises naturally from the Green's function normalization for the Laplace equation in 3D. (Ref: Ap).

Part 2: Isotropic Stress Normalization



The scalar pressure p is $\frac{1}{3}$ the trace of the stress tensor ($p = -\frac{1}{3} \sigma_{ii}$). The interaction couples to the full isotropic stress. The effective volume for pressure-work is $V_{eff} = 3 * V_{core} = 3 * \left(\frac{4}{3} \pi r_c^3\right) = 4\pi r_c^3$. (Ref: Appendix A).

These two effects combine to fix the ' 4π ' prefactor without continuous tuning.

The Bypass: Deriving Planck's Constant

Planck's constant is not a fundamental postulate in SST. It is a derived quantity fixed by the mechanical coupling A_C and a dimensionless kinematic ratio, α .

Standard Approach

Postulate h

SST Derivation

Input 1: Dimensionless
Kinematic Ratio α
 $\alpha \equiv 2\|\mathbf{v}_C\|/c$.
(Value: $7.297... \times 10^{-3}$)

Input 2: Mechanical
Coupling $A(\text{SST})_C$
(Value: $2.307... \times 10^{-28} \text{ J m}$)

Connection
 $\alpha = \frac{A_C}{\hbar c}$

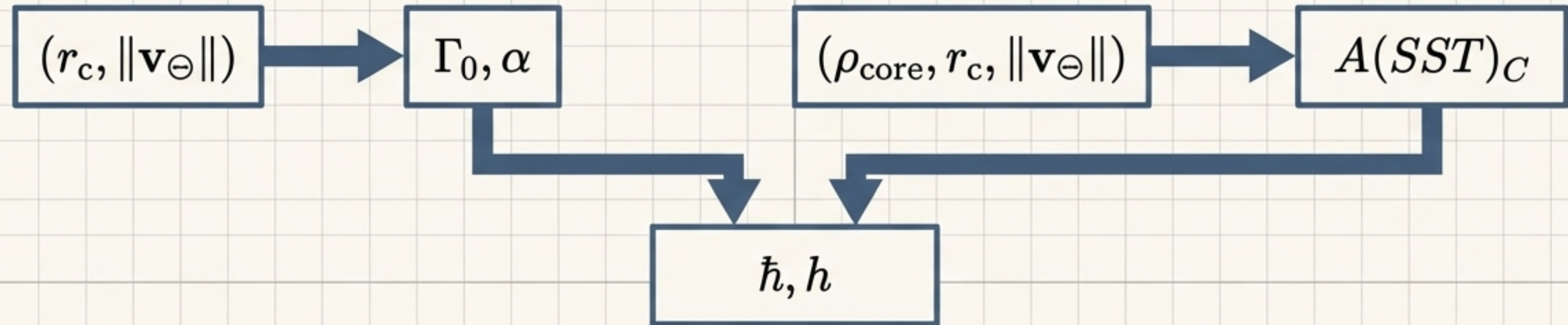
Derivation
 $\hbar = \frac{A(\text{SST})_C}{\alpha c}$

Derived $\hbar$: $1.0545717167 \times 10^{-34} \text{ J s}$

Derived h : $6.6260695157 \times 10^{-34} \text{ J s}$

These values are obtained with zero free parameters, given the initial triad.

The Complete Parameter-Free Derivation Chain



Quantity	Formula	Value (SST)
Primitives (Inputs)		$\rho_f, r_c, \ \mathbf{v}_\Theta\ , \rho_{\text{core}}$ values
α	$2\ \mathbf{v}_\Theta\ /c$	7.297×10^{-3}
$A_C^{(\text{SST})}$	$4\pi\rho_{\text{core}}\ \mathbf{v}_\Theta\ ^2 r_c^4$	$2.307 \times 10^{-28} \text{ J m}$
$\hbar$	$A_C^{(\text{SST})}/(\alpha c)$	$1.055 \times 10^{-34} \text{ J s}$
h	$2\pi\hbar$	$6.626 \times 10^{-34} \text{ J s}$

The Mediator: An Effective Field Theory for the Pressure Perturbation

The interaction is mediated by a scalar field ϕ , where the physical pressure perturbation δp is proportional to ϕ . A minimal quadratic EFT Lagrangian naturally describes its dynamics.

$$\left(\frac{1}{c_\phi^2} \partial_t^2 - \nabla^2 + m_\phi^2 \right) \phi = \frac{J}{Z}.$$

Case 1: Massless Mediator ($m_\phi \rightarrow 0$)

$$\nabla^2 \delta p = 0$$

Potential: $V(r) \propto \frac{1}{r}$ (Unscreened Monopole)

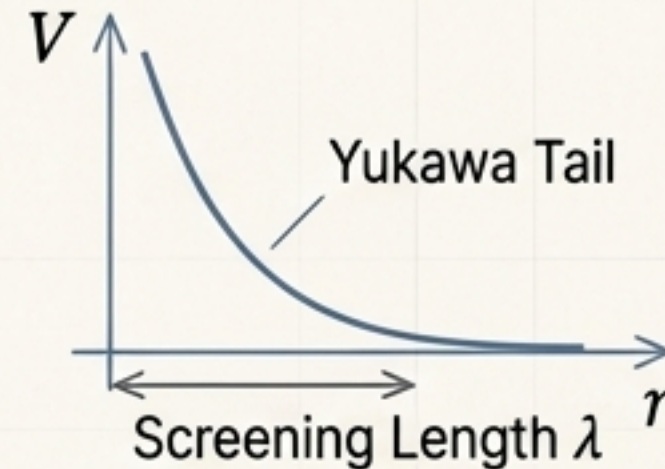


Status: This is the channel used to derive A_C .

Case 2: Massive Mediator ($m_\phi > 0$)

$$(\nabla^2 - \lambda^{-2}) \delta p = 0, \text{ where } \lambda = m_\phi^{-1}$$

Potential: $V(r) \propto e^{-r/\lambda}$ (Yukawa Tail)



Status: A natural, falsifiable alternative if long-range suppression is observed.

The EFT provides a concrete carrier (ϕ -waves) for energy and momentum, underpinning the 'echo' prediction.

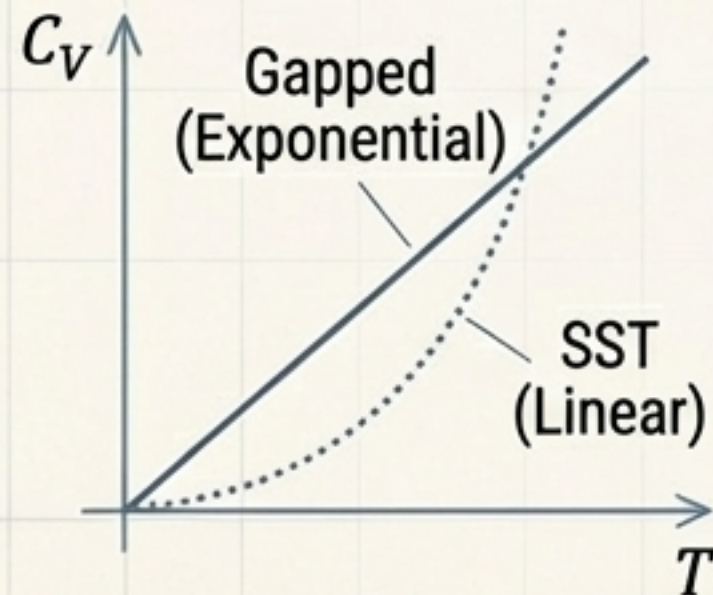
Falsifiability as a Virtue: Testable Predictions

SST is not just a mathematical construct; it makes specific, testable predictions that distinguish it from other models.

Low-Temperature Heat Capacity

SST Predicts: Linear scaling, $C_V \propto T_{\text{swirl}}$, due to one-dimensional gapless modes on the filament.

Falsifier: Observation of exponentially suppressed heat capacity, which would imply a gapped spectrum.

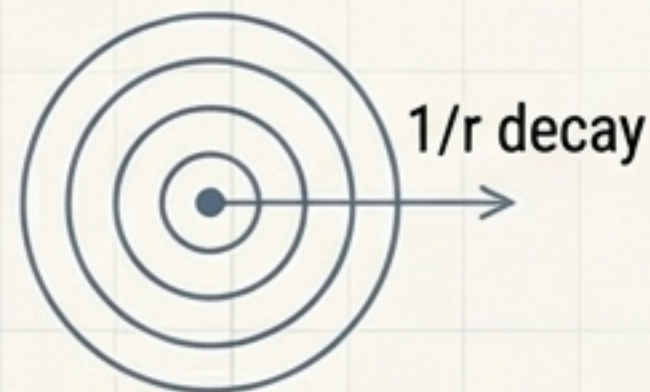


Far-Field Interaction Scaling

SST Predicts: A dominant $1/r$ monopole potential.

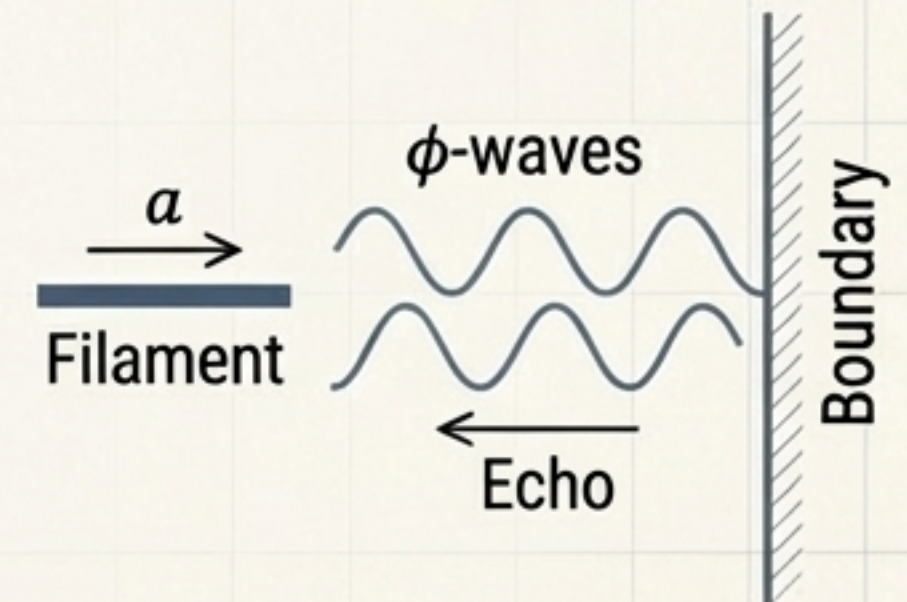
$$\lim_{r \rightarrow \infty} |r * V(r)| = \text{const.}$$

Falsifier: Robust observation of a $1/r^2$ or faster decay, indicating the absence of the monopole channel and forcing a revision to a higher-multipole model. This also directly tests for a finite screening length λ .



Acceleration-Induced Echo Channel

SST Predicts: An accelerating filament should generate propagating pressure/foliation waves (ϕ -waves) that can be detected as a delayed "echo." This is a hydrodynamic analog to the Unruh effect.



The Remaining Architecture: The Mass Selection Problem

While $\hbar$ is derived, the masses of particles remain an open question. SST provides a new structural context for this problem.

Structural Identity

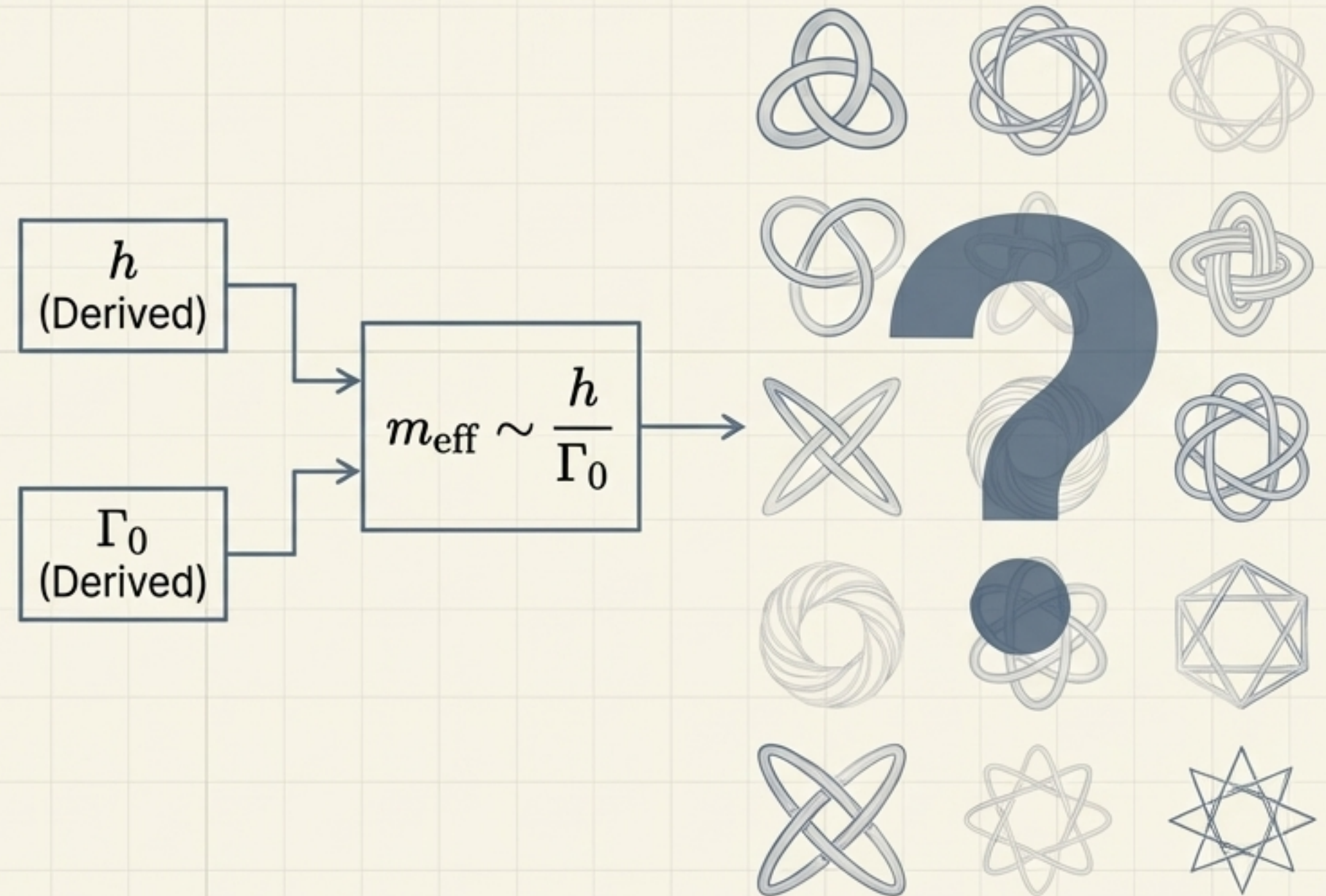
The theory retains the structural form of circulation quantization found in superfluids:

$$\hbar \sim m_{\text{eff}} * \Gamma_0$$

The Re-Framed Question

Since $\hbar$ and Γ_0 are now derived from the primitive inputs, this identity can be rearranged to define an effective mass: $m_{\text{eff}} \sim \hbar / \Gamma_0$

The Selection Problem: Which specific filament topology (knot type, geometry) or thermodynamic state corresponds to a stable configuration with this m_{eff} , thereby selecting the observed particle mass spectrum?



The Swirl-String Theory Edifice: A Summary

Foundation (The Postulates)

- Inviscid, incompressible hydrodynamic medium.
- Minimal primitive triad: $\{\rho_f, r_c, \|v_\ominus\|\}$.
- Clausius-consistent work/heat decomposition.

Core Derivations (The Structure)

- A parameter-free derivation of the coupling constant A_C from mechanical inputs.
- A bypass of the Planck postulate, deriving $\hbar$ from A_C and the kinematic ratio $\alpha = 2\|v_\ominus\|/c$.
- A coherent EFT framework for the interaction mediator.

Outlook (The Vista)

- A set of specific, falsifiable predictions (heat capacity, force law scaling, echo channel).
- A well-defined next step: solving the mass selection problem.

SST presents a logically complete, zero-free-parameter chain from classical hydrodynamics and thermodynamics to the constants of quantum mechanics.